RL Order Zone Trading Strategy

Architecture, Entry Logic, Trade Management & Exit Rules

Instrument: ES / NQ Micro Futures | Timeframe: 5-Minute Bars

1. Strategy Overview

This strategy uses a Recurrent PPO (LSTM) reinforcement learning agent to trade ES/NQ micro futures on 5-minute bars. The agent is trained to enter positions only at high-confluence Order Zones — price areas where supply/demand imbalance, liquidity sweeps, and rejection candles align simultaneously.

The agent operates within a fully defined Markov Decision Process (MDP): one episode equals one RTH session (~78 bars). At each bar it selects from five possible actions, subject to hard safety masks.

Core Design Philosophy

- Risk is defined BEFORE entry — stop is placed beyond the zone, target is a function of remaining ATR.
- R-multiples are the unit of measurement — P&L, rewards, and all thresholds are denominated in R.
- Process quality is rewarded — the agent learns to prefer high-confluence setups, not just profitable ones.
- Hard guardrails cannot be overridden by the agent — masking enforces non-negotiable risk rules.

2. Action Space (5 Actions)

At every 5-minute bar the agent selects exactly one of the following actions:

Action	Code	Description
HOLD / WAIT	0	Do nothing — stay flat or hold current position without change
ENTER SHORT	1	Open a short position at current bar close price
ENTER LONG	2	Open a long position at current bar close price
EXIT	3	Close the current position immediately at market price
TRAIL STOP	4	Activate / advance the trailing stop to protect open profits

Actions are masked before the agent samples — invalid actions are removed from the probability distribution entirely, so the agent never 'tries' a forbidden action.

3. Entry Logic

3.1 Hard Entry Masks (absolute blocks)

These conditions completely prevent entry — they cannot be overridden by the agent:

Condition	Rule
Position already open	Only one trade open at a time — no pyramiding
Max trades per day reached	Default: 10 trades per session maximum
Loss streak pause	After 3 consecutive losses, entries blocked for 6 bars
End of session	No new entries in the last 3 bars of RTH
Max drawdown breached	If daily loss ≥ 3R, only HOLD is allowed

3.2 Entry Quality Scoring (via Reward Function)

When entry masks allow an entry, the agent is incentivised to enter at high-quality setups through bonuses and penalties applied immediately at entry:

Component	Value (R)	Trigger
Full Order Zone Confluence	+0.50	All three pillars present: zone + liquidity + rejection
In Supply/Demand Zone	+0.20	Price inside a valid S/D zone
Liquidity Sweep Present	+0.15	Recent liquidity sweep detected
Rejection Candle Present	+0.10	Pin bar or engulfing candle confirmed
ATR Has Room	+0.15	ATR not yet in warning/exhaustion territory
High R:R Ratio	+0.20	Setup R:R ≥ minimum threshold
Entry Cost	−0.005	Transaction cost proxy — applied on every entry
No Zone Present	−0.05	Entry attempted with no valid zone
ATR Exhausted	−0.40	Session range already used ≥ 95% of daily ATR
R:R Below Minimum	−0.08	Risk/reward ratio below 1.5:1
No Liquidity Confluence	−0.05	No liquidity sweep or sweep evidence
Overtrading	−0.20	Entry after daily trade limit approached

3.3 Three Pillars of a Valid Order Zone Entry

An Order Zone requires all three pillars to receive the full +0.50 confluence bonus:

- **Pillar 1:** Supply/Demand Zone — A consolidation followed by an impulse move. The zone is the consolidation area. Zones expire after 200 bars or 3 touches.
- **Pillar 2:** Liquidity Sweep — Price runs a swing high/low (clearing stop orders) then reverses. The wick must be ≥ 3% of ATR.
- **Pillar 3:** Rejection Candle — Either a pin bar (wick ratio ≥ 2×) or an engulfing candle (body ratio ≥ 1.1×) forming at the zone boundary.

3.4 Stop Loss Placement

Stop loss is placed beyond the zone boundary, not a fixed number of points:

- Method: Zone-based — stop placed just outside the consolidation area
- Buffer: 3% of daily ATR beyond the zone top (long) or zone bottom (short)
- Rule: Stop can NEVER be widened after entry (hard rule in position manager)
- Example — Long: zone base at 5200, ATR = 100 pts → stop at 5200 − (100 × 0.03) = 5197

3.5 Take Profit Calculation

Take profit is dynamically sized using remaining ATR at the time of entry:

- Method: ATR-remaining — target = 75% of the daily ATR not yet consumed by the session range
- Minimum R:R: 1.5:1 required (risk/reward must be at least 1.5 before entry is scored favourably)
- Example: ATR = 100 pts, session range used 40 pts, remaining = 60 pts → target distance = 60 × 0.75 = 45 pts

4. Trade Management While in Position

4.1 Position Sizing

Contracts are dynamically sized to risk a fixed percentage of account equity per trade:

Parameter	Value	Description
Risk per trade	1% of account	Fixed fractional sizing
Min contracts	1	Floor — always at least 1 contract
Max contracts	3	Cap — never more than 3 contracts
Point value	$2.00 (MES/MNQ)	Dollar value per point per contract

Formula: n_contracts = floor(account × 0.01 / (stop_distance_pts × point_value)), clamped to [1, 3].

4.2 Trailing Stop Logic

The agent can activate a trailing stop by selecting the TRAIL STOP action. Once active, the stop advances automatically every bar the action is maintained.

Stage	Trigger	Behaviour
Activation	Unrealised P&L ≥ 2R AND agent selects TRAIL STOP	Trailing stop activates; initial trail distance = ATR × 25%
Normal trail	Each bar agent selects TRAIL STOP	Stop ratchets up (long) / down (short) — never reverses
Aggressive	Unrealised P&L ≥ 4R	Trail tightens further — agent rewarded for selecting trail here
Locked in	Trail activates	Minimum 2R is locked in — stop will never fall below entry + 2R

TRAIL STOP is masked (unavailable) when unrealised P&L < 2R — the agent cannot trail prematurely and risk tightening the stop before the trade has room.

4.3 Same-Bar Ambiguity Rule

When a single candle's range touches BOTH the stop loss and take profit simultaneously, the strategy always assumes the STOP was hit first (worst-case fill). This prevents the agent exploiting ambiguous candles as guaranteed take-profits.

- Long: if bar low ≤ stop AND bar high ≥ target → stop loss (loss recorded)
- Short: if bar high ≥ stop AND bar low ≤ target → stop loss (loss recorded)

5. Exit Conditions (Priority Order)

Exits are processed in strict priority order each bar:

Priority	Exit Reason	Trigger	Exit Price
1 (highest)	Stop Loss	Bar low ≤ stop (long) or bar high ≥ stop (short)	Stop price
1 (highest)	Trailing Stop	Bar crosses trailing stop level	Trailing stop price
2	Take Profit	Bar high ≥ target (long) or bar low ≤ target (short)	Target price
3	Agent Exit	Agent selects EXIT action	Current bar close
4	Session End	Last bar of RTH session reached with open position	Bar close
5	Max Drawdown	Total daily loss (realised + unrealised) ≥ 3R	Market close

Exit Rewards and Penalties

Component	Value (R)	Trigger
Trailing stop correctly used	+0.30	Position exited via trailing stop
Aggressive trail at 4R	+0.20	Trail activated with unrealised ≥ 4R
Exit at ATR target	+0.20	Take profit hit at the ATR-based target
Held past 4R without trail	−0.50	Position held beyond 4R unrealised without trailing
Gave back profit	−variable	Weight × R given back from peak unrealised

6. Risk Guardrails

Guardrail	Limit	Consequence
Daily loss limit	−3R	Trading stops for the session; only HOLD allowed
Max trades per day	10	Entry actions masked for remainder of session
Loss streak pause	3 consecutive	Entry blocked for 6 bars; then unblocked
No late entries	Last 3 bars	No new entries in last 15 minutes of RTH
ATR exhaustion gate	≥ 95% of ATR	Entry penalised −0.40R (soft, not hard mask)
Max drawdown per episode	5R total	Episode terminates immediately
Max drawdown breach penalty	−2.00R	Hard violation reward applied
Daily loss limit breach	−1.00R	Hard violation reward applied

7. Reward Function Summary

Core Reward

The primary reward signal on a closed trade is the R-multiple outcome scaled by the agent's running win rate:

$$\text{core_reward} = \text{pnl_r} \times \text{win_rate}$$

Win rate starts at 0.5 (Laplace prior) before any trades and updates after each closed trade. This means the agent is rewarded not just for profitable trades, but for being consistently profitable — a single lucky win is discounted.

Hold-Flat Penalty

A penalty of −0.025R is applied every bar the agent holds flat (no open position). This is critical for trade frequency: the cumulative daily cost of staying flat (≈78 bars × 0.025 = 1.95R/day) exceeds the expected cost of a typical loss, forcing the agent to trade 4–5 times per day rather than staying idle.

Reward Component Hierarchy

Category	Components	Purpose
Core outcome	pnl_r × win_rate	Primary learning signal
Entry shaping	Zone/liquidity/rejection bonuses & penalties	Learn WHERE to enter
Exit shaping	Trail bonuses, gave-back penalties	Learn WHEN and HOW to exit
Discipline	Hold-flat penalty, overtrading penalty	Learn HOW OFTEN to trade
Violations	Drawdown/loss-limit breach penalties	Enforce risk guardrails

8. Model Architecture

Component	Detail
Algorithm	Recurrent PPO (RecurrentPPO) with LSTM — from sb3_contrib
Memory	LSTM hidden state persists across bars within an episode
Action masking	MaskablePPO-style masking applied before sampling
Parallelism	16 SubprocVecEnv workers during training
Normalisation	VecNormalize on observations (not rewards) — stats saved per checkpoint
Entropy annealing	Entropy coefficient: 0.05 (start) → 0.005 (end) over 1M steps
Total timesteps	2,000,000 per fold
Eval frequency	Every 50,000 steps

Checkpoint Saving — Tiered Phase Gates

Models are saved only when both conditions are met: (1) composite score ≥ phase threshold, and (2) ≥ 8 trades evaluated (prevents saving lucky zero-trade runs).

Phase	Step Range	Min Composite Score Required
0 — Warmup	0 – 400k	0.05
1 — Post-warmup	400k – 750k	0.15
2 — Consistency	750k – 1.1M	0.25
3 — Quality	1.1M – 1.5M	0.35
4 — Elite	1.5M+	0.45

Composite Score Formula

Composite score (0–1) used for model ranking and save gates:

Metric	Weight	Reference ('Perfect') Value	Direction
Annualised Sharpe Ratio	30%	1.5	Higher = better
Total PnL (R)	25%	20R	Higher = better
Win/Loss Ratio	25%	3.0	Higher = better
Max Drawdown (R)	20%	8R	Lower = better

Note: Sharpe ratio is annualised by multiplying the episode-level ratio by $\sqrt{252}$. A Sharpe of 1.5 represents an excellent live strategy.

9. Walk-Forward Validation

To test robustness across different market regimes, training uses a rolling walk-forward framework:

Parameter	Value
Training window	252 trading days (≈ 12 months)
Validation window	26 trading days (≈ 5 weeks)
Fold shift	26 days — each fold shifts forward by one val window
Agent per fold	Fresh agent + fresh VecNormalize per fold (no leakage)
Output	Per-fold Excel journal + Plotly HTML equity curves

After training, test_fold.py loads all saved checkpoints, runs deterministic evaluation on the held-out test period, and outputs a ranked summary table with per-checkpoint Plotly candlestick charts showing entries, exits, SL/TP lines, and the equity curve.

10. Complete Trade Lifecycle

1. **Bar-by-bar scan:** Each 5-min bar: observation built from ATR state, zone detector, liquidity detector, trend classifier, order zone engine, and portfolio state (26-40 features).
2. **Action masking:** Hard safety masks applied — invalid actions removed from agent's choice set.
3. **Agent decision:** LSTM processes observation + hidden state → selects action from masked distribution.
4. **Entry execution (if ENTER LONG/SHORT):** Stop = zone boundary − ATR buffer. Target = entry + 75% remaining ATR. Contracts sized to risk exactly 1% of capital.
5. **In-trade management:** Each subsequent bar: agent can HOLD (do nothing), EXIT early, or TRAIL STOP. Stop/target checked against bar high/low — stop is priority if both hit.
6. **Trailing stop advancement:** When agent selects TRAIL STOP and unrealised ≥ 2R: stop ratchets behind price by ATR × 25%. Stop only moves in favour — never reversed.
7. **Exit triggered:** Priority: stop hit → take profit hit → agent EXIT → session end → max drawdown.
8. **Reward computed:** core_reward = pnl_r × win_rate, plus all shaping components. Reward stored for PPO update.
9. **Episode summary:** At session end: metrics computed (win rate, PnL, Sharpe, drawdown). Composite score checked against phase gate. Checkpoint saved if gate passes.

Appendix: Key Configuration Values

Parameter	Value	File
Bar timeframe	5 minutes	environment_config.yaml
Session type	RTH (09:30–16:00 ET)	environment_config.yaml
Initial account balance	$2,500	environment_config.yaml
Risk per trade	1% of account	risk_config.yaml
Min / Max contracts	1 / 3	risk_config.yaml
Stop method	Zone-based	risk_config.yaml
Zone buffer (ATR %)	3%	risk_config.yaml
Take profit method	75% remaining ATR	risk_config.yaml
Minimum R:R	1.5:1	risk_config.yaml
Trail activates at	2R unrealised	risk_config.yaml
Trail aggressive at	4R unrealised	risk_config.yaml
Trail lock-in	2R minimum	risk_config.yaml
Daily loss limit	3R	risk_config.yaml
Max trades/day	10	risk_config.yaml
Loss streak threshold	3 consecutive	risk_config.yaml
Hold-flat penalty	−0.025R / bar	reward_config.yaml
Max drawdown (episode)	5R	risk_config.yaml
ATR exhaustion gate	95% of daily ATR	risk_config.yaml
No-entry window	Last 3 bars of RTH	risk_config.yaml
Min trades to save model	8	trading_eval_callback.py